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Storytelling Through Skill Trees

Discussing observations in developmental psychology vs how development is presented in video game skill trees, and how those things can help us tell stories.

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18 Min Read

Humans are not trees. It's a radical statement to make, I know, but try to bear with me while I explain this difficult concept! Humans, in fact, do not begin as a single entity and then branch out to catch the sun's light. Instead, when we grow what we try to "catch" are various abilities, skills, and knowledge as a means of reaching a personal goal. No two of us are chasing the same sunlight.

Which is my really obnoxious way of saying: Why are we so convinced that skill trees are a great solution to map character growth over time? If we look at evolutionary psychology in books like "The Pocket Guide to Polyvagal Theory" or "The Strange Order of Things" we find that psychology professionals have identified that organisms basically grow in whatever way they find the most success. Even trees and plants will change their growth patterns over time based on the resistance that they meet in their surroundings. So if all of the biological world is based on the concept of adapting to their unique stressors, why are we building all of our games with rigid skill trees? Why does my D&D character sheet require that I be a wizard or a bard or a fighter? Why do I pick my difficulty when I begin the game, and get locked into these systems? This stuff doesn't fit our evolutionary needs. Adapt or die, these are the rules of the whole world. If your system doesn't allow adaptation, then it incentivizes death.

The common wisdom found in game design communities will, however, contradict me here. Players WANT to be a Bard because they don't want to have to do a ton of research into building their character for a silly game they play twice a month on tuesdays. Figuring out the entire system just to understand how to build the most powerful character is an awful way for most of society to spend their free time (obviously some people do like these things, and more power to you! Literally!). People want to walk down a path somewhat mindlessly, and make a few choices along the way knowing that they will still have a good time in the process. This is a natural desire for each of us, because we want to feel like we're good at things without having to spend our lives mastering those things. In fact, even in real life, most of us are unconcerned with becoming the most or the best, we're largely just trying to figure out how to Make The Thing in the first place. The idea of becoming the best Thing Maker alive is absurd.

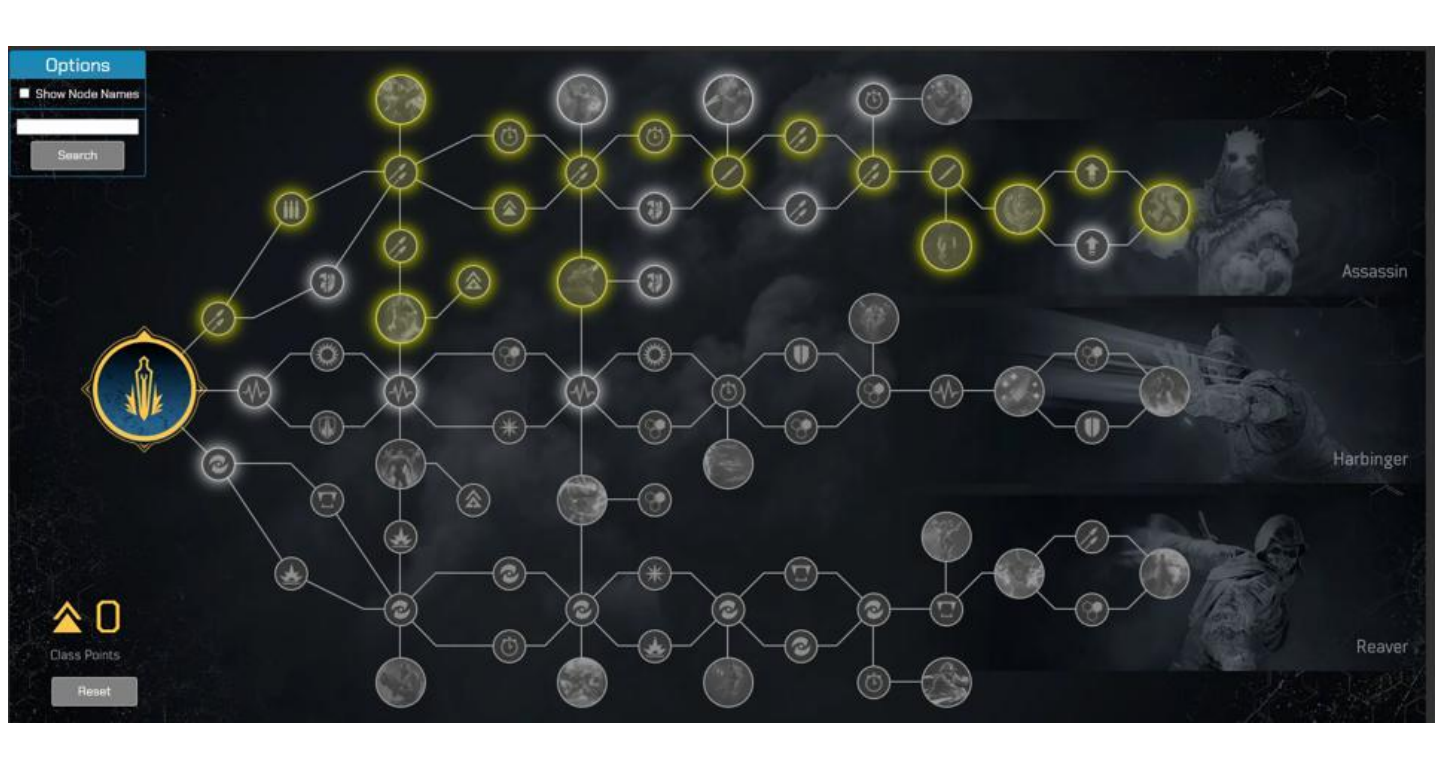
Fortunately for us, we as humans have already spent boatloads of time and energy learning how to educate other humans in healthy ways, and to do so while they are quite literally incapable of understanding how to plan out a complex skill tree path. As children, we are placed into schools which teach us a set curriculum, and as we advance through our education we pick up skills along the way. Some of us will pick up more math skills than others, some will pick up primarily physical talents, and still others will focus on the arts. Often, we will pick up a set of skills society enforces upon us, which we do not find daily use for. How often have you or your friends complained about having to take that advanced calculus class, knowing your career goal is to bake cupcakes? How often have you been out with a group of friends and you break out that "party trick" you learned years ago but never had any specific use for because you learned it completely aside from whatever your core path was? We all have those "useless" skills that don't come out as a daily occurrence because we thought they'd be worth it at the time.

The point that I'm making is that a person isn't a collection of skills carefully chosen, but a collection of skills of which only a carefully chosen few are repeated. In other words, I like to design games, and so I write a lot about game design, and so people view me as a game designer. I'm also an artist, but I do that much less, so people view me as an artist much less. If your Cleric doesn't also have a set of skills from when they tried being a Barbarian, are they even a real person?

WWBJD?

First grade teachers will cover a generic set of skills we think are essential. Reading, Writing, Basic Math, etc. are all represented. By the time you reach 18 you have dropped any number of skills by the wayside. But we treat characters in games as if they start at the beginnings of a tree, and branch out as they go. Reality tends to work in reverse, where we start in the branches and funnel into increasingly-specific direction over time. If school were a skill tree, you'd start off with an extremely nonlinear pile of skill to choose from, and somewhere at the half way mark of the game, you'd finally have the 30 required skills to earn your certification as a Paladin. Once you're a Paladin, you can officially dive into those deep Paladin skills that just would not be useful to anyone else. The "Paladin" class, in this example, acts as a signpost. It's a guide saying "if you come this direction, your experience will look like this" and is a good way to direct people towards a specific goal. This concept has been proven to work so well that every school system on the planet finds some rough approximation of it, even if the details vary. So why on earth do we build games the other way around? Why am I choosing my class at the start of my character? Why am I locked in with lateral movement a near impossibility? And look, I acknowledge that multiclassing in an RPG is there to solve exactly this, but come on folks, we all know that's not the same thing. In real life I don't get a degree in art and then a minor in english and then only get to either be an artist or a writer for the rest of my life. That fluidity of adaptation is essential for human evolution and happiness both, and is something we should strive to include in our game design. And ok, saying all this is well and good, but how do we make it happen?

The key features of this evolutionary look at personal growth is that we start off general and end up specific after walking through a series of gates. This really isn't unlike how Outriders handles its skill tree.



On the left we effectively have elementary education, then there's a clear delineation where you can branch into a new path if you choose. Unfortunately, you're stuck with only a few places where you can jump between the three trees in this image, so this works more like a D&D multiclass system where you can choose a new career path, but you're defined by the choice you made when accepting that first job (our company LOVES lateral movement among our employees, just fill out these 6 forms and reapply after a minimum of 2 years in your current role and you'll be considered for that pay cut!). The way humans like to work is to have a more open pool available. Mechanically, this would be more like drawing from a large pool of skills which get increasingly specific as you go along. For those familiar with CCG games, this is sort of like drafting a pack of Magic cards. You start with a broad pool of possibilities, but each choice you make narrows your focus down so that later picks become almost predetermined. Which sounds rather confusing, so let me make a chart to demonstrate that visually.

Let's avoid debating how to organize those DnD classes, that's not the point here. But the idea is the same regardless of how you structure things. On the left is a list of skills for everyone, and as you go right it filters down to be skills for each specific class. If we were to continue the Magic The Gathering example, the green box on the left would include cards of all kinds, then the next column represents when we're given boxes of cards for each specific color, then next would be deck types, and the red on the far right would be boxes full of cards catered to popular meta decks. As you move from left to right, you go from generic to extreme specificity, just as you do when you progress from elementary school to college. Your development is along whichever path you like, and then once you reach the end you must prove you have achieved mastery. Is your MTG deck a "red" deck if you have 5 red cards in it? 10? In real life, if you want to be known as skilled in a particular area, you take a certification exam for that skillset. In other words, you qualify yourself as "a red deck" by proving that you have enough skills to use the term. In this skill tree, you do the same thing. If you want to call yourself a rogue, you have to have… let's say 10 skills in the rogue tree. Once you have any 10 rogue skills, you are officially a rogue. Congrats! I don't care if you have 25 points in wizard, or 45 points in paladin, if you have 10 rogue points, you qualify to call yourself as a rogue.

This aligns much more closely to how we, as humans, expect to learn. There are goals to reach at the end of the path we've chosen, and it's our task to reach the goal we choose. No two people will learn in exactly the same way, but generally you get to where you want to go. As opposed to the purely linear skill trees that we see in games like Outriders, or the ones we get in games like Path of Exile which start off at a particular class and then try to add the nonlinearity at the end, which is the opposite of how we humans learn and grow.

There is, of course, quite a good reason people tend to design games in this "backwards" way: People get overwhelmed by too many options. If you start off in the "Everyone" bucket, you have far too many choices at once, and how on earth could you ever possibly choose? Again, we look at developmental psychology for our answer. We don't allow kids to choose the classes they take, we give them a set curriculum and they learn all of it. Once a kid knows more of what they like, we start to give them choices to make, such as choosing types of curriculum to take in high school (which sometimes takes the form of choosing blocks of classes, and other times in choosing entire high schools). Then we don't let them fully branch out until college, once we know what they like. So in the above example, perhaps you could choose a class to start with, and then change that class at any time. If you choose, for example, Wizard, you would begin collecting skills within the wizard curriculum, moving from generic skills into magical skills, and if you reach magical skills and suddenly realize you want to pivot into a Warlock instead, well that's fine because you haven't reached the point where the skills required is different from a Warlock anyway, so no big deal. Once you choose Warlock as your new class, your current skill list is the same, you just have a new goal. If you choose to then become a Fighter, well you'll have to start over from the physical skills list instead of the magical skills list, but at least you don't have to relearn the skills for everyone! And already this feels much more like what it's like to be a person going through a real education process, realizing that you want to learn something new.

The other aspect of this is what to do with unused skills, and to which I suggest limiting the immediate skill list you have, such as how you can only build a deck of 60 cards in Magic The Gathering. Then you take all the skills you've learned, apply the "build" you want, and there's your character. This is also how we work in real life, we've all got those party tricks we can break out, but our core "build" is just the skills we use over and over in our daily life.

This guy gets it

But this isn't an article about game design, despite how much time I've devoted to that so far. No, this is an article about narrative design. Mapping skill trees to human development patterns doesn't add all that much to the game design process, but it fundamentally changes how we are allowed to tell stories in these games. Imagine you're the child of strict parents who are forcing you into Paladin college, but you've always known you're a Bard in your heart. Maybe you take night classes and dream of running off to the circus, maybe you only look onto Bard skills from a distance, but either way you have a problem: You don't WANT to progress the way you have been. Now your skill tree is greyed out, and there's only one path to choose. You can see the other roads, but they only bring you agony. You can never be anything but what you are. Life is awful.

At least life was awful, until you met Him. He changed you. He showed you the way things could be, he showed you the world. He taught you a Bard skill! It was a simple one, but you have a skill point on that forbidden tree! How is that possible? Weren't those other trees greyed out? Were they? Or could you have put those skills there at any time, if only you'd allowed yourself? Sure, your parents will be pissed now, but it's not like they can stop you from learning, right? And maybe they'll even tolerate a few skills, I mean it's not like you're at the real hard stuff yet, right? It takes a while before you can unlock those Bard-specific skills anyway, this is just a phase, you'll grow out of it before then. Which, of course, you don't. You get your first Bard skill and you've crossed the threshold. Your parents see this trigger and disown you. They know you have a Bard skill, and no son of mine will be a filthy Bard! Not under MY roof!

In this example we have a set of skills that unlock based on triggers reached by hitting a threshold. Then we have quests that react to those same triggers, your parents changing their dialogue based on if the boolean variable "hasBardSkill" has been flipped to true. It's simple stuff, programmatically. What's important, though, is that we have that natural movement from unskilled to skilled which matches human development. If you started off by choosing to be Paladin, and only ever got the chance to multiclass, this storytelling would be impossible. You'd already be hard-coded into your class from day one with no way out, and what is a story without personal growth and change over time?

But I'm not leaving it there either, because I drew the earlier connection to developmental psychology for a reason. Characters in stories have goals. In some cases, those goals have to do with reaching a particular location on a skill tree. In other cases, they may just want to be a different person in a more abstract sense. Maybe the character is a child who just really wants to be like Daddy, so they try to learn the same skills he learned. Maybe the person has anger management issues and needs to develop those skills, or maybe they've found themselves trapped in a life of crime and they want out. Maybe they are getting bullied and the only reason they care about the skill tree mess is to keep them from being hurt. People grow over time, at all stages of life. There is a biological imperative in all of us which tells us to keep adapting, even if many of us try to ignore it because we don't want to believe that we can change, or because we're simply happy where we are. In games we map this growth onto what we call a Skill Tree, but in reality what we're mapping is the development of our own minds. If a person has a mental block against allowing themselves to become a particular type of person, something common among abuse victims, that might manifest as a greyed-out skill tree, as in my earlier example. A skill tree is nothing more than the path of growth a character is allowed to move along. If the path is linear, then that person is only capable of growth in a particular direction. If the path is open, the person is allowed to grow more freely. Which shape the tree exists in will determine the possible reality that will define the person. Or at least the reality they impose upon themself. What kind of person has a linear skill tree? What does that say about their political or religious beliefs? What kind of career might they prefer?

To offer an example, consider the film Easy A. It's a romantic comedy where the protagonist feels invisible as a teenage girl in high school. Her goal is to be popular, and the moment she trips that "isPopular" trigger, her life completely spirals out of control. Sunk cost fallacy keeps her there for a while, but eventually she has to come clean and resume her normal life. She's not trying to become a Wizard or a Barbarian, but she is definitely trying to learn a new set of skills in the Popular Girl tree. Those skills, however, cause her nothing but trouble and her story focuses on how she must quickly learn to adapt. As another example, look at the anime Mob Psycho 100. Mob is a psychic high schooler who must avoid using his powerful psychic skill, which is triggered by stress. He has the skill unlocked already, but wants a new set of skills over on the "Physical Skills" branch instead. He has found that the more physical skills he has unlocked, the more able he is to control his use of those psychic skills he's so worried about. It's all a great way of mapping depression onto verb use, and shows us how we can demonstrate a character's personality based on what skills they have, and which ones they feel like they need. These are two examples, but just consider the stories you've read or watched. Consider how the protagonist uses their skills in a way that either helps or hurts their progress through the story. Consider what skills they have at their disposal and why they did or did not use those skills in that story. Try to imagine what their skill tree might look like.

Each of us has a skill tree full of nodes we've unlocked. Each of us has a set of skills we are willing to use, and a set of skills we are unwilling to use. If we were a videogame, our preferences and the walls we build for ourselves would all manifest as variations in the UI that presents our stats to the person controlling us.

What class are you? What class do you not consider a possibility for yourself? Is it because that tree is in some way locked for you, or is it simply that you'd have to go too far back on your tree to switch to that other path you'd need to be on?

Hopefully that gives everyone something to think about. Thanks for reading! I'd love your comments and thoughts below!

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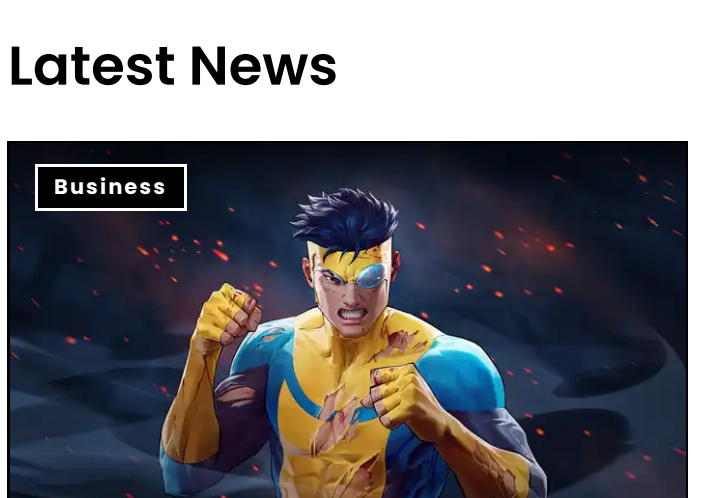

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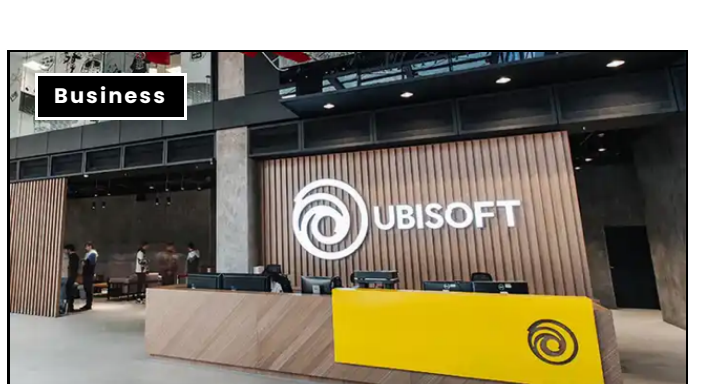

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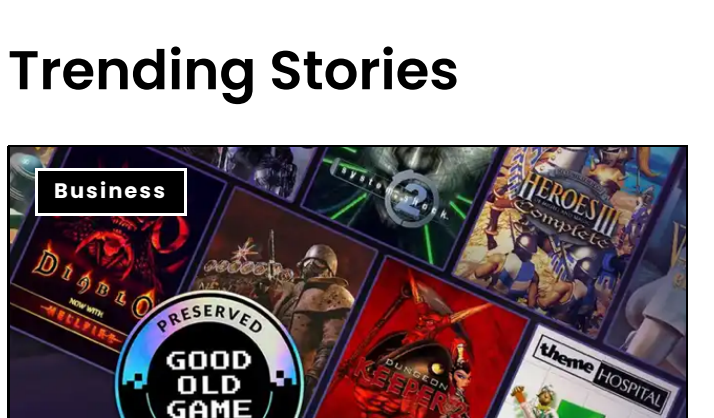

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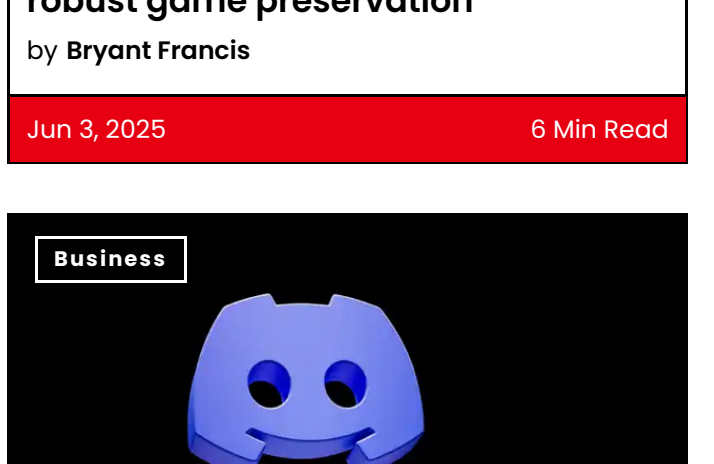

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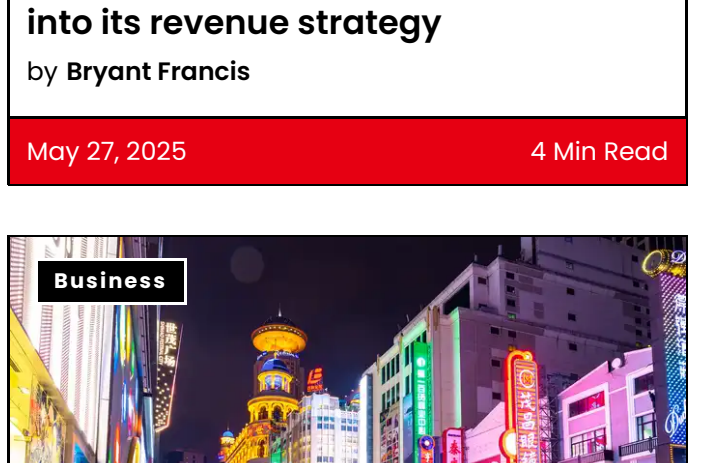

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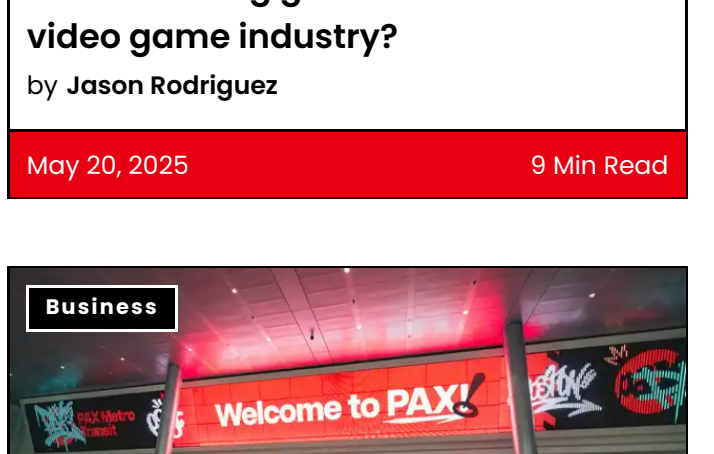

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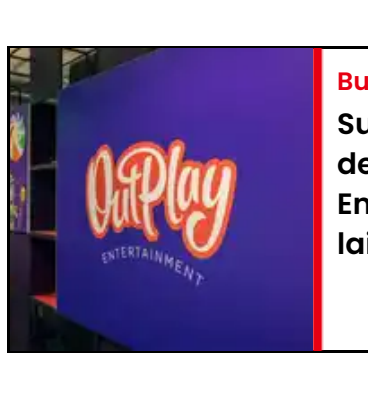

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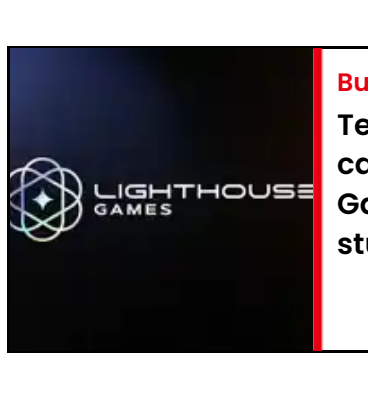

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